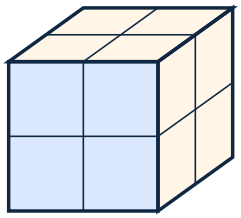
Volume with Fractions

Fill the box with small cubes. 8 small cubes make 1.

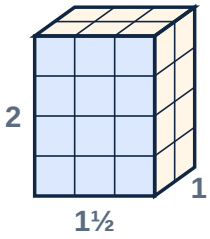
Small cubes: ½ unit on every edge



2 across, 2 back, 2 up = 8 small cubes.
Together they make 1 unit cube.

8 small cubes = 1 cubic unit

A box 1½ by 1 by 2



Small cubes: 3 × 2 × 4 = 24.
24 ÷ 8 = 3 unit cubes.

$$1\frac{1}{2} \times 1 \times 2 = 3$$

Length × width × height works with fractions too.

Your turn!

$$2\frac{1}{2} \times 2 \times 2 = \boxed{10}$$

For teachers

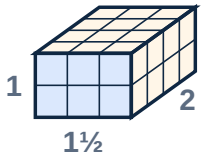
Goal: volume of boxes with fraction edges, by packing ½-unit cubes and by l × w × h (6.G.2). Say “How many small cubes? 8 small cubes make one.” Then check with l × w × h.

Volume with Fractions: Practice

Count small cubes, then multiply the edges.

Part 1 Count the small cubes Each small cube is ½ on every edge. 8 make 1 cubic unit.

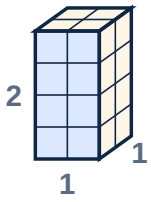
1)



24 small cubes

24 ÷ 8 = 3 cubic units

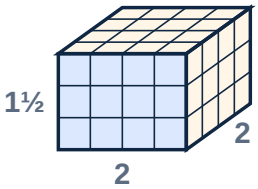
2)



16 small cubes

16 ÷ 8 = 2 cubic units

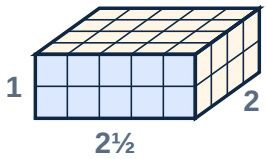
3)



48 small cubes

48 ÷ 8 = 6 cubic units

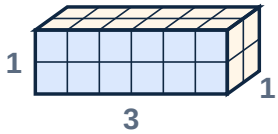
4)



40 small cubes

40 ÷ 8 = 5 cubic units

5)



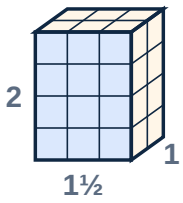
24 small cubes

24 ÷ 8 = 3 cubic units

Volume with Fractions: Practice (continued)

Part 1 Count the small cubes (continued)

6)



24 small cubes

24 ÷ 8 = 3 cubic units

Part 2 Length × width × height Multiply the fractions.

7) $2\frac{1}{2} \times 4 \times 3 = 30$

8) $1\frac{1}{2} \times 2 \times 5 = 15$

9) $\frac{1}{2} \times 6 \times 5 = 15$

10) $3\frac{1}{2} \times 2 \times 2 = 14$

11) $\frac{3}{4} \times 4 \times 6 = 18$

12) $2\frac{1}{2} \times 1\frac{1}{2} \times 4 = 15$

13) $4\frac{1}{2} \times 2 \times 3 = 27$

14) $\frac{1}{3} \times 6 \times 9 = 18$

15) $1\frac{1}{2} \times 1\frac{1}{2} \times 4 = 9$

16) $1\frac{1}{4} \times 8 \times 2 = 20$

Volume with Fractions: Practice (continued)

Part 3 Answers with fractions The answer is a mixed number.

17) $1\frac{1}{2} \times 3 \times 1 = 4\frac{1}{2}$

18) $2\frac{1}{2} \times 3 \times 1 = 7\frac{1}{2}$

19) $\frac{1}{2} \times 7 \times 3 = 10\frac{1}{2}$

20) $3\frac{1}{2} \times 3 \times 1 = 10\frac{1}{2}$

21) $1\frac{1}{2} \times 5 \times 3 = 22\frac{1}{2}$

22) $\frac{1}{2} \times \frac{1}{2} \times 6 = 1\frac{1}{2}$

23) $2\frac{1}{2} \times 5 \times 1 = 12\frac{1}{2}$

24) $\frac{1}{4} \times 5 \times 3 = 3\frac{3}{4}$

Part 4 Base × height The base is already worked out.

25) $7\frac{1}{2} \times 4 = 30$

26) $12 \times 2\frac{1}{2} = 30$

27) $4\frac{1}{2} \times 6 = 27$

28) $10 \times 3\frac{1}{2} = 35$

Volume with Fractions: Practice (continued)

Part 4 Base × height (continued)

29) $10\frac{1}{2} \times 2 = \boxed{21}$

30) $8 \times 2\frac{1}{4} = \boxed{18}$

Part 5 Find the missing edge Volume ÷ the other two edges.

31) $4 \times 2\frac{1}{2} \times \boxed{3} = 30$

32) $6 \times 1\frac{1}{2} \times \boxed{2} = 18$

33) $\frac{1}{2} \times 8 \times \boxed{5} = 20$

34) $5 \times 3\frac{1}{2} \times \boxed{2} = 35$

35) $1\frac{1}{2} \times 4 \times \boxed{6} = 36$

36) $2 \times 4\frac{1}{2} \times \boxed{4} = 36$

Part 6 More length × width × height

37) $4\frac{1}{2} \times 7 \times 4 = \boxed{126}$

38) $2\frac{1}{2} \times 7 \times 4 = \boxed{70}$

39) $1\frac{3}{4} \times 3 \times 4 = \boxed{21}$

40) $\frac{1}{3} \times 4 \times 3 = \boxed{4}$

Volume with Fractions: Practice (continued)

Part 6 More length × width × height (continued)

41) $1\frac{2}{3} \times 4 \times 3 =$ **20**

42) $1\frac{1}{3} \times 8 \times 3 =$ **32**

43) $1\frac{1}{2} \times 7 \times 4 =$ **42**

44) $4\frac{1}{2} \times 7 \times 6 =$ **189**

45) $2\frac{1}{2} \times 5 \times 6 =$ **75**

46) $\frac{3}{4} \times 7 \times 4 =$ **21**

47) $2\frac{1}{4} \times 4 \times 4 =$ **36**

48) $\frac{2}{3} \times 6 \times 2 =$ **8**

49) $2\frac{1}{2} \times 6 \times 1 =$ **15**

50) $\frac{1}{2} \times 2 \times 3 =$ **3**

51) $\frac{3}{4} \times 8 \times 3 =$ **18**

52) $\frac{1}{2} \times 8 \times 5 =$ **20**

Volume with Fractions: Practice (continued)

Part 7 More base × height

53) $4\frac{1}{2} \times 5 = 22\frac{1}{2}$

54) $1\frac{1}{2} \times 2 = 3$

55) $4\frac{1}{2} \times 2 = 9$

56) $3\frac{1}{2} \times 5 = 17\frac{1}{2}$

57) $11\frac{1}{2} \times 5 = 57\frac{1}{2}$

58) $9\frac{1}{2} \times 7 = 66\frac{1}{2}$

59) $8\frac{1}{2} \times 4 = 34$

60) $7\frac{1}{2} \times 8 = 60$

61) $5\frac{1}{2} \times 2 = 11$

62) $2\frac{1}{2} \times 2 = 5$

Volume with Fractions: Practice (continued)

Part 8 More missing edges

63) $6 \times 4\frac{1}{2} \times \boxed{2} = 54$

64) $7 \times 4\frac{1}{2} \times \boxed{6} = 189$

65) $3 \times 4\frac{1}{2} \times \boxed{4} = 54$

66) $6 \times 4\frac{1}{2} \times \boxed{3} = 81$

67) $5 \times 2\frac{1}{2} \times \boxed{2} = 25$

68) $4 \times \frac{1}{2} \times \boxed{2} = 4$

69) $5 \times 4\frac{1}{2} \times \boxed{2} = 45$

70) $3 \times 1\frac{1}{2} \times \boxed{2} = 9$

Part 9 Remember? Problems from earlier packets.

71) $16 \times 5 \div 2 = \boxed{40}$

72) $x + 26 = 61$: $x = \boxed{35}$

73) $\frac{5}{6} \div \frac{1}{12} = \boxed{10}$

74) 25% of 64 = $\boxed{16}$

Volume with Fractions: Practice (continued)

Part 9 Remember? (continued)

75) lcm of 6 and 10 = 30

76) 7.2 ÷ 0.8 = 9

Part 10 Word problems Length × width × height.

77) A box is 3 ft long, 2 ft wide and 1½ ft tall. What is its volume?

Number sentence 3 × 2 × 1½ = 9

Answer 9 cubic ft

Draw or show your work

78) A tank is 5 m by 2 m. Water fills it ½ m deep. How much water is that?

Number sentence 5 × 2 × ½ = 5

Answer 5 cubic m

Draw or show your work

79) A box holds small cubes 4 across, 6 back and 2 up, each ½ inch. How many cubic inches?

Number sentence 4 × 6 × 2 = 48, 48 ÷ 8 = 6

Answer 6 cubic in

Draw or show your work

Volume with Fractions: Practice (continued)

Part 10 Word problems (continued)

80) A box has a base of 12 sq cm and holds 30 cubic cm. How tall is it?

Number sentence $30 \div 12 = 2\frac{1}{2}$

Answer **$2\frac{1}{2}$** cm

Draw or show your work

81) Challenge: A drawer is 2½ ft long, 2 ft wide and ½ ft deep. Two of them: what volume in all?

Number sentence $2\frac{1}{2} \times 2 \times \frac{1}{2} \times 2 = 5$

Answer **5** cubic ft

Draw or show your work

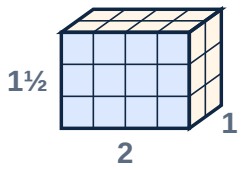
Check-In: Volume with Fractions

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Count the small cubes

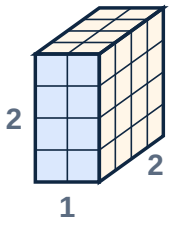
1)



24 small cubes

24 ÷ 8 = 3 cubic units

2)



32 small cubes

32 ÷ 8 = 4 cubic units

Part 2 Length × width × height

3) $\frac{1}{2} \times 8 \times 3 = 12$

4) $3\frac{1}{2} \times 4 \times 2 = 28$

5) $2\frac{1}{2} \times 3 \times 3 = 22\frac{1}{2}$

6) $1\frac{1}{2} \times 4 \times 3 = 18$

Part 3 Find the missing edge

7) $3 \times 2\frac{1}{2} \times 4 = 30$

8) $1\frac{1}{2} \times 6 \times 2 = 18$